S.B.JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Session-2025-26 (Even)

TAE-I Project Based Learning

Risk Assessment System

Submitted By

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AI Technique(s) (2)	Problem Representation (4)	Tools Explored (2)	Outcome (2)	Total (10)
Sign Of Student:			Sign Of Course In-Charge	

1. Industry Problem Context

1. Industry Problem Context

In today's world, risk assessment plays a very important role in many industries such as banking, insurance, cybersecurity, healthcare, and e-commerce. Organizations must evaluate risk before making important decisions like approving loans, issuing insurance policies, detecting fraud transactions, or allowing access to secure systems.

Risk assessment is the process of identifying, analyzing, and evaluating risk based on different factors such as financial data, user behavior, transaction history, login attempts, and other important parameters. Traditionally, risk assessment is done manually by experts, which is time-consuming, slow, and sometimes inaccurate due to human error.

To solve this problem, industries use Artificial Intelligence-based Risk Assessment Systems that automatically calculate risk scores using mathematical and probability-based methods. These systems analyze input data, calculate the probability of risk, and classify the risk into categories such as Low Risk, Medium Risk, and High Risk. This helps organizations make fast, accurate, and consistent decisions.

The Risk Assessment System is widely used in:

- Banking systems for loan approval risk
- Insurance companies for policy risk calculation
- Cybersecurity systems for user risk detection
- E-commerce websites for fraud transaction detection
- Healthcare systems for patient risk prediction

Therefore, developing a Risk Assessment System using probability theory is an important AI application that helps industries in decision-making, risk prediction, and risk management.

2. Feasibility in IT Systems

The Risk Assessment System is highly feasible in IT systems because it uses mathematical probability and simple computational methods to calculate risk scores. The system can be easily developed using Python and can run on basic computer systems without requiring high-end hardware.

In this system, user or system data is taken as input, such as income, credit score, transaction amount, login attempts, or other risk-related factors. Each factor is assigned a probability value between 0 and 1 based on its risk level. The system then applies a weighted probability formula to calculate the overall risk score. Based on the final score, the system classifies the risk into Low Risk, Medium Risk, or High Risk.

The system is feasible in IT environments for the following reasons:

- The system uses simple mathematical calculations, so processing is fast.
- Python programming language makes implementation easy and efficient.
- The system requires very low memory and processing power.
- It can be implemented as a desktop application or web application.
- The system can be integrated with databases to store user data and risk history.
- The system can be used in real-time risk analysis systems.

Because of these reasons, the Risk Assessment System is practical, cost-effective, and suitable for real-world IT applications such as banking systems, cybersecurity systems, insurance systems, and fraud detection systems.

3. AI Techniques Used in Industry

1. Probability Theory

Probability theory is the main AI technique used in this system. It is used to calculate the likelihood of risk based on different input factors. Each risk factor is assigned a probability value between 0 and 1, where 0 represents low risk and 1 represents high risk. The system then calculates the overall risk score using a weighted probability formula.
Probability theory is widely used in:

- Banking risk analysis
- Insurance risk calculation
- Fraud detection systems
- Cybersecurity risk analysis
- Medical risk prediction

2. Rule-Based Classification

Rule-based classification is used to classify the calculated risk score into different categories such as Low Risk, Medium Risk, and High Risk. The system uses predefined rules to classify risk.
Example:

- If Risk Score < 0.3 → Low Risk
- If Risk Score between 0.3 and 0.7 → Medium Risk
- If Risk Score > 0.7 → High Risk

This technique is commonly used in expert systems and decision support systems.

3. **Predictive Analysis**

Predictive analysis is used to predict future risk based on current and past data. The system analyzes input data and predicts whether the user or transaction is risky or safe.
Predictive analysis is used in:

- Loan approval prediction
- Credit scoring systems
- Intrusion detection systems
- Fraud detection
- Healthcare risk prediction

These AI techniques make the Risk Assessment System intelligent, fast, accurate, and useful for real-world industry applications.

1. Representation/Modeling in Industry Systems

In this project, the Risk Assessment System is modeled using a probability-based risk model. The system is designed by dividing it into Input, Processing, Risk Model, and Output. This structured representation helps in understanding how the system calculates and predicts risk.

1. Input Representation

The system takes various risk-related factors as input. These inputs represent the parameters that affect the risk level.

Examples of input factors:

- Age
- Income
- Credit Score
- Loan Amount
- Transaction Amount
- Number of Failed Login Attempts
- User Activity / Behavior

Each input is converted into a risk value between 0 and 1, where:

- 0 represents Low Risk
- 1 represents High Risk

These inputs are stored as variables in the system.

2. Risk Model Representation

The system uses a weighted probability model to calculate the overall risk score. Each risk factor is assigned a weight based on its importance.

The risk score is calculated using the weighted average probability formula:

Risk Score = (w1 × F1 + w2 × F2 + w3 × F3 + … + wn × Fn) / (w1 + w2 + w3 + … + wn)

Where:

- w = weight of each factor
- F = risk value of each factor
- n = number of risk factors

This model is called a **Probability-Based Risk Model**.

3. Processing Model

The system processes data in the following steps:

1. Collect input data from the user or database.
2. Convert input data into probability risk values (0 to 1).
3. Assign weights to each risk factor.
4. Apply weighted probability formula.
5. Calculate final risk score.
6. Classify risk level.

4. Output Representation

The system produces the following output:

- Final Risk Score (between 0 and 1)
- Risk Category:
  - Low Risk
  - Medium Risk
  - High Risk

The output can be displayed in:

- Text format
- Table format
- Graph format

This representation model is similar to real-world risk assessment systems used in banking, insurance, and cybersecurity industries.

3. Tools & Technologies in Industry

The Risk Assessment System is developed using tools and technologies that are commonl used in Artificial Intelligence and data analysis applications. These tools help in dataprocessing, risk calculation, and result visualization.

Programming Language

Python

- Python is used to develop the Risk Assessment System.
- It is easy to learn and widely used in AI and data science applications.
- Python supports mathematical calculations, data processing, and machine learning.

Libraries Used

1. **NumPy**
   - Used for mathematical and probability calculations.
   - Helps in calculating weighted risk scores.
2. **Pandas**
   - Used for data handling and data processing.
   - Helps in storing and analyzing risk data.
3. **Matplotlib**
   - Used for data visualization.
   - Helps in displaying risk score graphs and charts.
4. **Flask (Optional)**
   - Used to develop a web-based Risk Assessment System.
   - Helps in creating a user interface for input and output.

Development Environment

- VS Code
- Cursor AI
- PyCharm

System Requirements

- Basic Computer System
- Minimum 4GB RAM
- Python Installed
- Internet Connection (Optional)
- Operating System: Windows / Linux / Mac

These tools and technologies make the Risk Assessment System efficient, easy to implement, and suitable for real-world industry applications.

4. Industry-Level Outcome

The Risk Assessment System demonstrates how Artificial Intelligence techniques can be used to calculate and predict risk in real-world applications. The system provides useful outcomes similar to industry-level AI systems used in banking, insurance, cybersecurity, and fraud detection.

Automation

The system automatically calculates the risk score based on input factors without human intervention.

- Reduces manual work
- Saves time
- Increases productivity

Efficiency

The system performs risk calculations quickly using probability formulas.

- Faster than manual risk calculation
- Can be used in real-time systems
- Handles large amounts of data

Accuracy

The system uses mathematical and probability-based calculations to determine risk.

- Reduces human errors
- Provides accurate risk scores
- Improves decision-making

Consistency

The system follows the same rules and formulas every time.

- Produces consistent results
- Reliable system
- No variation due to human factors

Scalability

The system can be expanded in the future:

- Can include more risk factors
- Can be connected to databases
- Can be converted into a web application
- Can be integrated into banking or cybersecurity systems

Real-World Benefits

The Risk Assessment System can be used in:

- Banking systems for loan risk assessment
- Insurance companies for policy risk calculation
- Cybersecurity systems for user risk detection
- E-commerce systems for fraud detection
- Healthcare systems for patient risk prediction

Thus, the Risk Assessment System provides automation, accuracy, efficiency, and intelligent decision-making, which are important outcomes in real-world industry applications.

5.Real-World Industry Example/use

The Risk Assessment System is used in many real-world industries where decision-making depends on risk calculation and prediction. The system helps organizations identify whether a situation is safe or risky before taking action.

1. Banking Industry

In banking systems, risk assessment is used for loan approval and credit scoring.

- The system checks income, credit score, loan amount, and financial history.
- It calculates the risk score of the customer.
- If the risk is high, the loan may be rejected.
- If the risk is low, the loan may be approved.

2. Insurance Industry

Insurance companies use risk assessment to calculate policy risk.

- The system analyzes age, health condition, lifestyle, and past records.
- High-risk customers have higher insurance premiums.
- Low-risk customers have lower premiums.

3. Cybersecurity Industry

In cybersecurity, risk assessment is used to detect risky users and attacks.

- The system monitors login attempts, user behavior, and system access.
- If unusual activity is detected, the system marks it as high risk.
- This helps in preventing cyber attacks and unauthorized access.

4. E-commerce Industry

E-commerce companies use risk assessment to detect fraud transactions.

- The system checks transaction amount, location, and user behavior.
- If the transaction risk is high, the system blocks the transaction.

5. Healthcare Industry

In healthcare, risk assessment is used to predict patient health risk.

- The system analyzes medical history, age, and health parameters.
- It predicts disease risk and helps doctors in decision-making.

These examples show that the Risk Assessment System is an important AI application used in many real-world industries for risk prediction, decision-making, and safety management.

6. Implementation & Result

Implementation:

The Risk Assessment System takes the following inputs from the user to calculate the risk score:

1. Probability of Event (0–1)

- This represents the chance that a risk event will occur.
- Value is between 0 and 1.
- **Example:**
  - 0.10 = 10% chance
  - 0.30 = 30% chance
  - 0.80 = 80% chance

2. Impact (1–10)

- This represents how serious the damage will be if the risk occurs.
- Value is between 1 and 10.
- Higher value means higher damage.

Impact Value	Meaning
1–3	Low Impact
4–7	Medium Impact
8–10	High Impact

3. Exposure (1–10)

- This represents how often the system is exposed to risk.
- Value is between 1 and 10.
- Higher value means more exposure to risk.

Vulnerability Value	Meaning
1–3	Low Vulnerability
4–7	Medium Vulnerability
8–10	High Vulnerability

Risk Calculation Formula

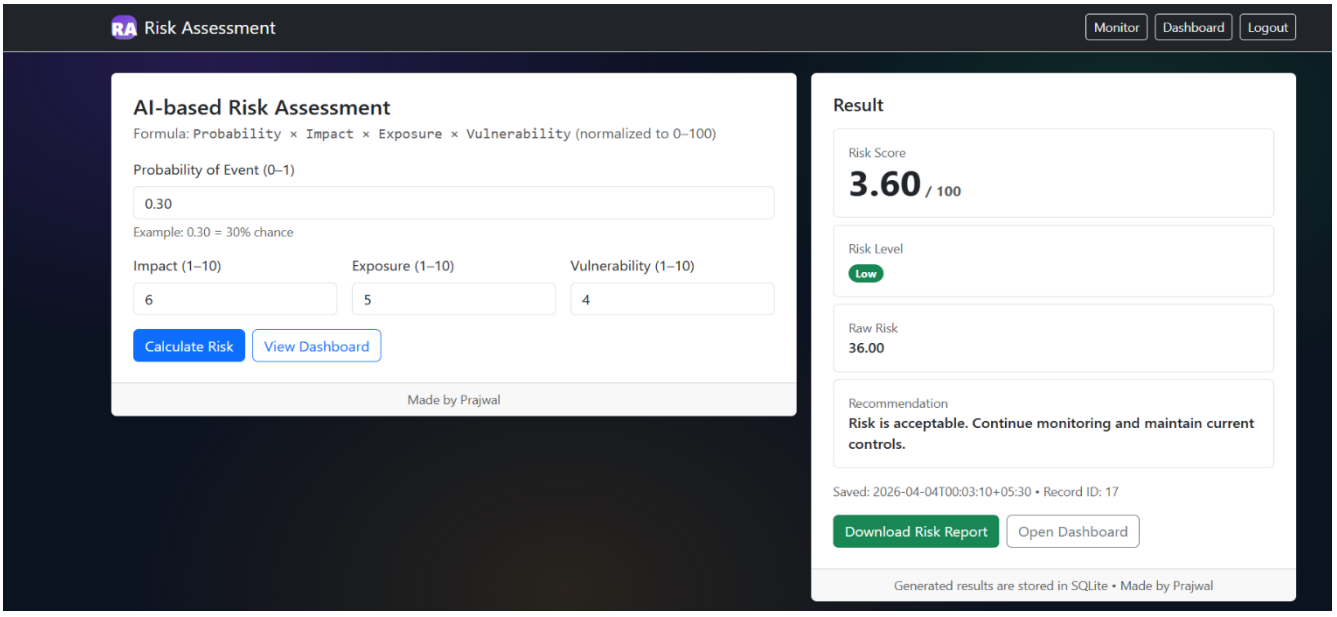
Risk Score = Probability × Impact × Exposure × Vulnerability

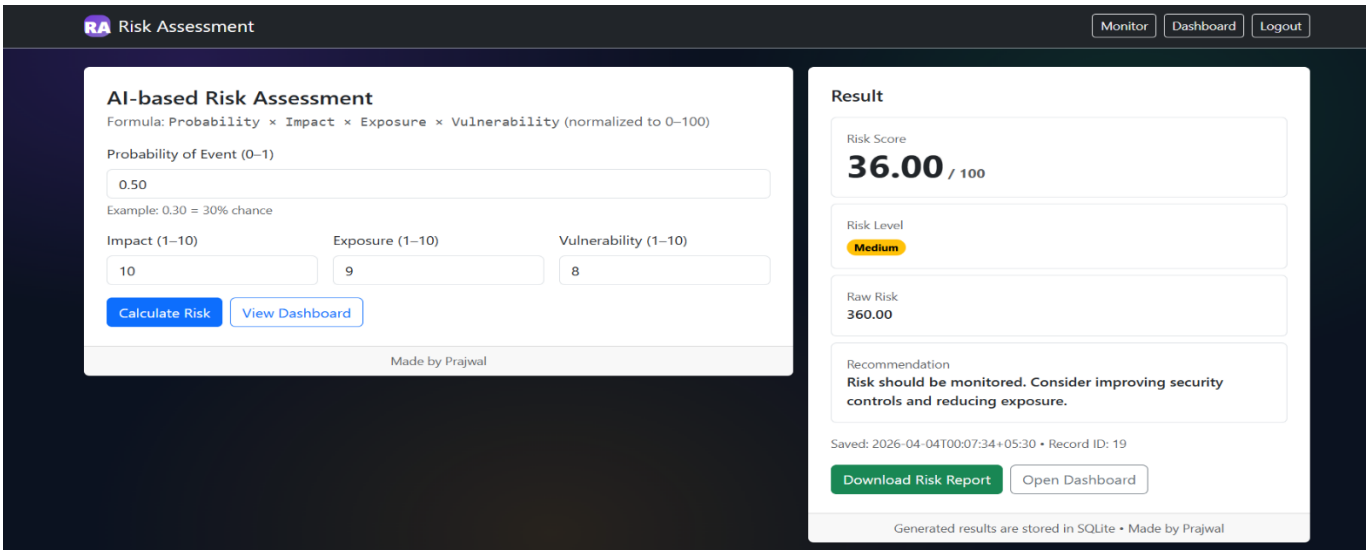
Then the result is normalized to 0–100.

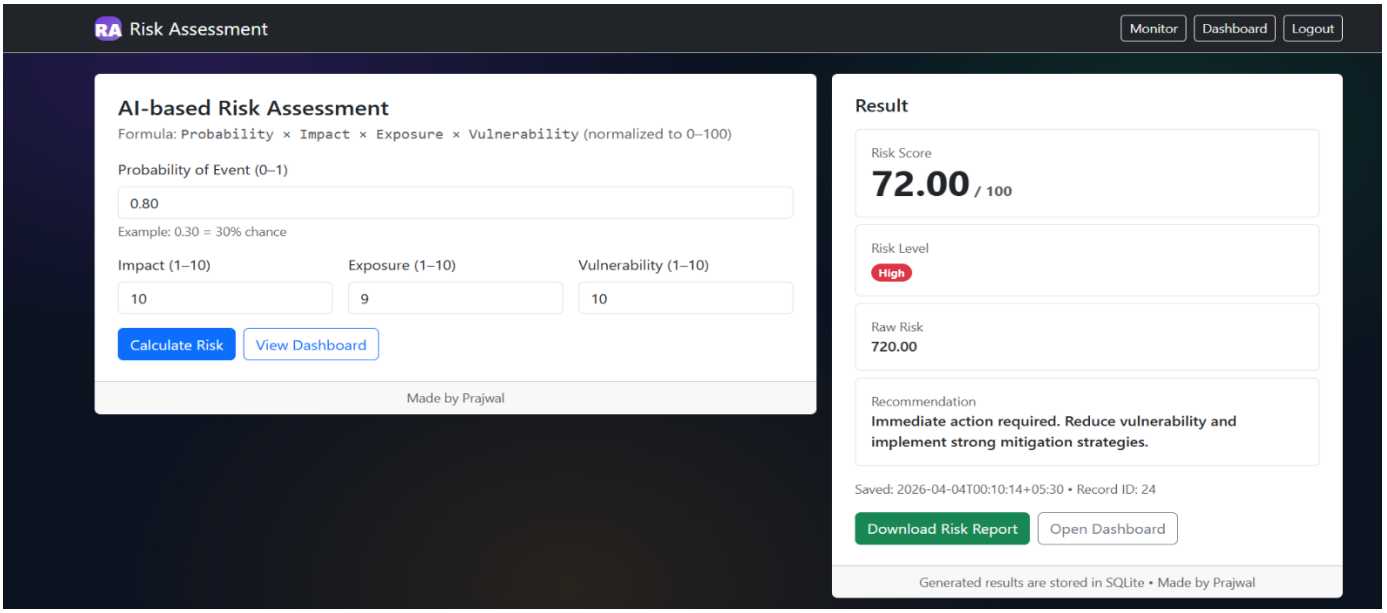
Example :

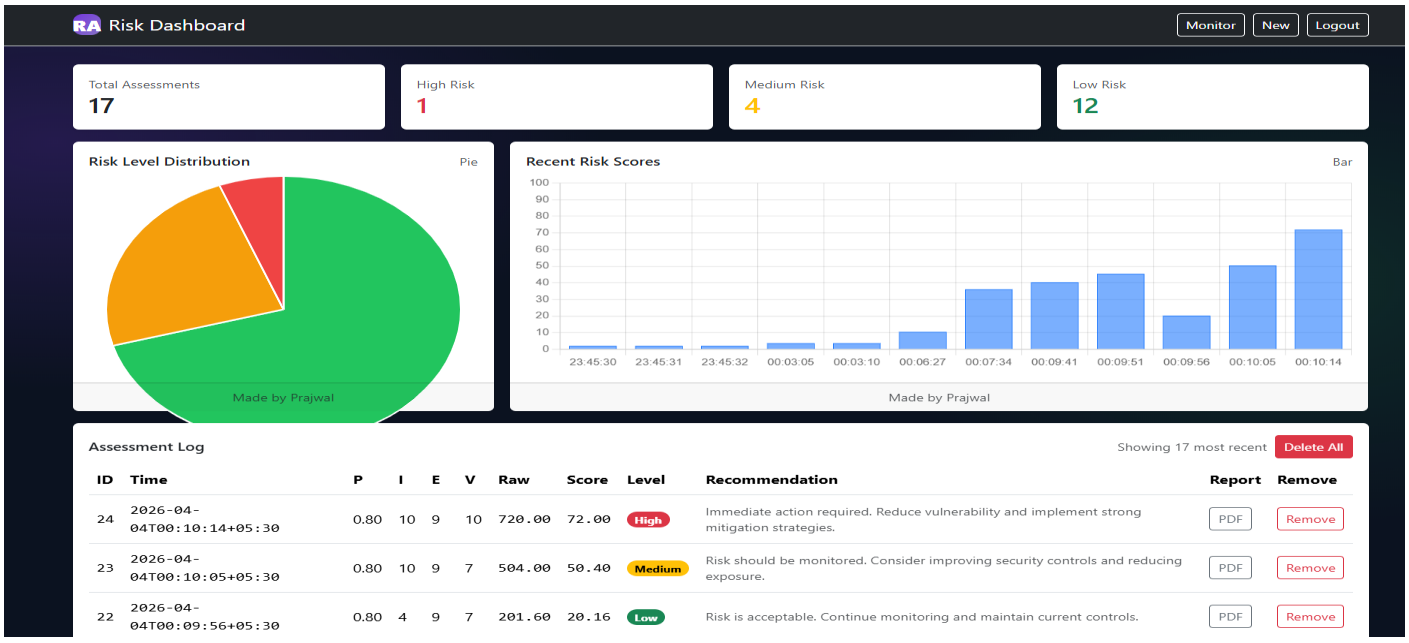
- **Probability** = 0.30
- **Impact** = 6
- **Exposure** = 5
- **Vulnerability** = 4

Raw Risk = 0.30 × 6 × 5 × 4 = 36
Risk Score = 3.60 / 100
Risk Level = Low




Risk Assessment
Monitor
Dashboard
Logout
AI-based Risk Assessment
Formula: Probability × Impact × Exposure × Vulnerability (normalized to 0–100)
Probability of Event (0–1)
0.50
Example: 0.30 = 30% chance
Impact (1–10)
10
Exposure (1–10)
9
Vulnerability (1–10)
8
Calculate Risk
View Dashboard
Made by Prajwal
Result
Risk Score
36.00 / 100
Risk Level
Medium
Raw Risk
360.00
Recommendation
Risk should be monitored. Consider improving security controls and reducing exposure.
Saved: 2026-04-04T00:07:34+05:30 • Record ID: 19
Download Risk Report
Open Dashboard
Generated results are stored in SQLite • Made by Prajwal


Risk Assessment
Monitor
Dashboard
Logout
AI-based Risk Assessment
Formula: Probability × Impact × Exposure × Vulnerability (normalized to 0–100)
Probability of Event (0–1)
0.80
Example: 0.30 = 30% chance
Impact (1–10)
10
Exposure (1–10)
9
Vulnerability (1–10)
10
Calculate Risk
View Dashboard
Made by Prajwal
Result
Risk Score
72.00 / 100
Risk Level
High
Raw Risk
720.00
Recommendation
Immediate action required. Reduce vulnerability and implement strong mitigation strategies.
Saved: 2026-04-04T00:10:14+05:30 • Record ID: 24
Download Risk Report
Open Dashboard
Generated results are stored in SQLite • Made by Prajwal


Risk Dashboard
Monitor
New
Logout
Total Assessments
17
High Risk
1
Medium Risk
4
Low Risk
12
Risk Level Distribution
Pie
Made by Prajwal
Recent Risk Scores
Bar
100
90
80
70
60
50
40
30
20
10
0
23:45:30
23:45:31
23:45:32
00:03:05
00:03:10
00:06:27
00:07:34
00:09:41
00:09:51
00:09:56
00:10:05
00:10:14
Made by Prajwal
Assessment Log
Showing 17 most recent
Delete All
ID Time P I E V Raw Score Level Recommendation Report Remove
24 2026-04-04T00:10:14+05:30 0.80 10 9 10 720.00 72.00 High Immediate action required. Reduce vulnerability and implement strong mitigation strategies. PDF Remove
23 2026-04-04T00:10:05+05:30 0.80 10 9 7 504.00 50.40 Medium Risk should be monitored. Consider improving security controls and reducing exposure. PDF Remove
22 2026-04-04T00:09:56+05:30 0.80 4 9 7 201.60 20.16 Low Risk is acceptable. Continue monitoring and maintain current controls. PDF Remove

7. Result

The Risk Assessment System was successfully implemented using Python and a probability-based risk calculation model. The system takes input parameters such as Probability, Impact, Exposure, and Vulnerability from the user and calculates the risk score using the risk formula. The calculated risk score is then normalized to a scale of 0 to 100 and classified into Low Risk, Medium Risk, or High Risk categories.

The system was tested with different input values to verify the results. For low input values, the system produced a Low Risk result. For moderate input values, the system produced a Medium Risk result. For high input values, the system produced a High Risk result. The system also generated recommendations based on the risk level, such as accepting the risk, monitoring the risk, or taking mitigation actions.

All the calculated risk records were successfully stored in the SQLite database, and the dashboard displayed previous risk records and risk analysis graphs. The system also generated a downloadable risk report for documentation purposes.

8. References

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